BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

The vessels and the rules of flow, measuring a blood pressure, how arterioles and capillaries do their jobs, venous return, and the reflexes and hormones that hold pressure steady. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

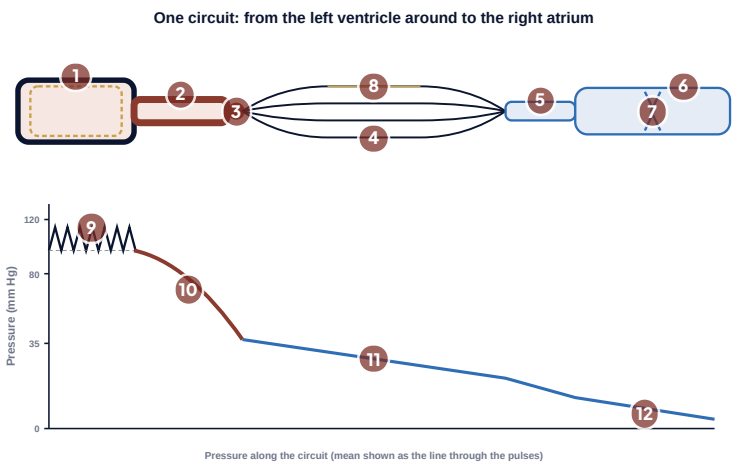
Name _____

A. The vessels, and the rules of flow

YOU SHOULD BE ABLE TO

Vessel structure and function. Match arteries, arterioles, capillaries, venules, and veins to their role in pressure buffering, resistance, exchange, and capacitance.

Pressure flow and resistance. Apply the relationship among flow, pressure gradient, and resistance and predict the effect of a change in vessel radius, blood viscosity, or vessel length on flow.



One circuit, with the pressure in each vessel

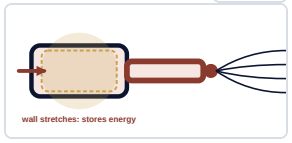
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WHAT YOU ARE LOOKING AT

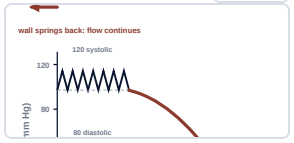
One lap of the systemic circuit, left to right: artery, arteriole, capillaries, venule, vein. The graph underneath is the pressure in each vessel. Wall thickness is drawn to tell you what each vessel is built to do.

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

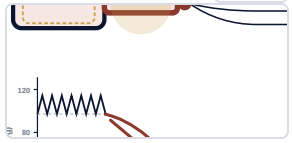
STEP 1



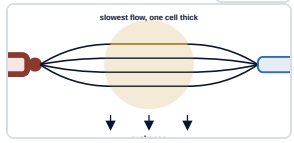
STEP 2



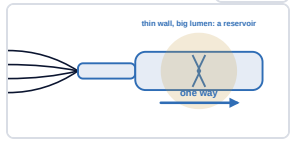
STEP 3



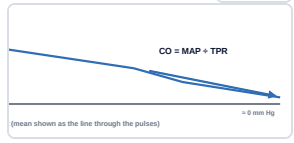
STEP 4



STEP 5



STEP 6



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

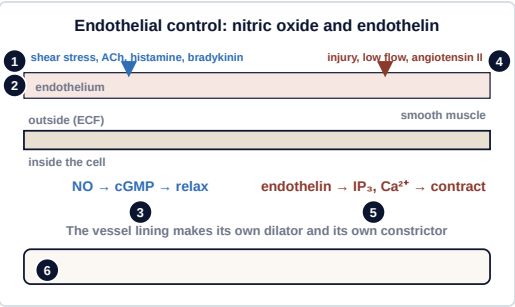
BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

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Name _____

A. The vessels, and the rules of flow (continued)

8 Endothelium: how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

THE RULES OF FLOW

Flow =

Resistance depends on

Halve the radius and

Double the viscosity (polycythemia) and

Length matters when

Velocity of flow is

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

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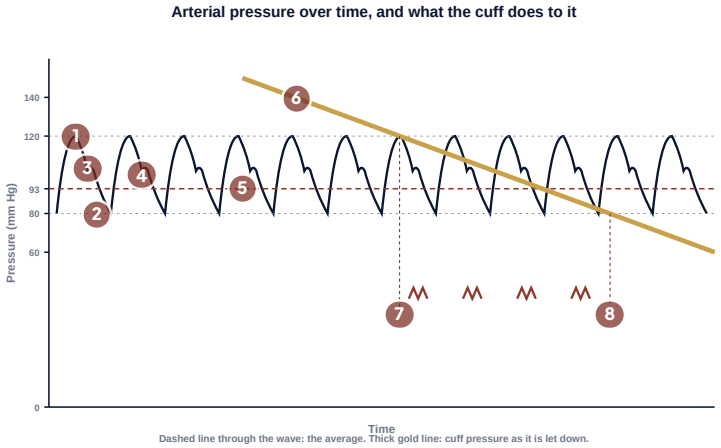
Name _____

B. Arterial blood pressure, and how you measure it

YOU SHOULD BE ABLE TO

Arterial blood pressure. Define systolic, diastolic, pulse, and mean arterial pressure, calculate mean arterial pressure, and explain how it is determined by cardiac output and total peripheral resistance.

Blood pressure measurement. Measure blood pressure by auscultation, explain the origin of the Korotkoff sounds, and record the response to a postural or exercise challenge.



The arterial pressure wave and the cuff

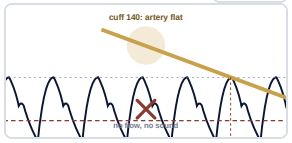
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WHAT YOU ARE LOOKING AT

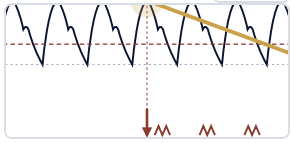
Left: the pressure in a large artery, beat by beat. Right: the same wave with a cuff pressure (gold) being let down across it. The sounds you hear through the stethoscope are drawn below the wave.

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

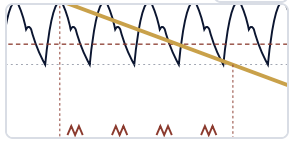
STEP 1



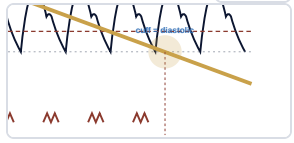
STEP 2



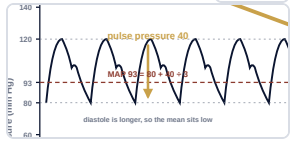
STEP 3



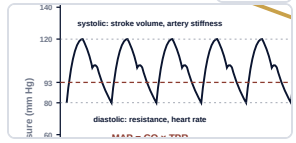
STEP 4



STEP 5



STEP 6



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

B. Arterial blood pressure, and how you measure it (continued)

WORK THE NUMBERS

Pulse pressure = systolic − diastolic

MAP = diastolic + ⅓ pulse pressure

MAP = cardiac output × total peripheral resistance

A patient at 150/70

A patient at 100/85

Posture or exercise in lab

Extra notes and drawings

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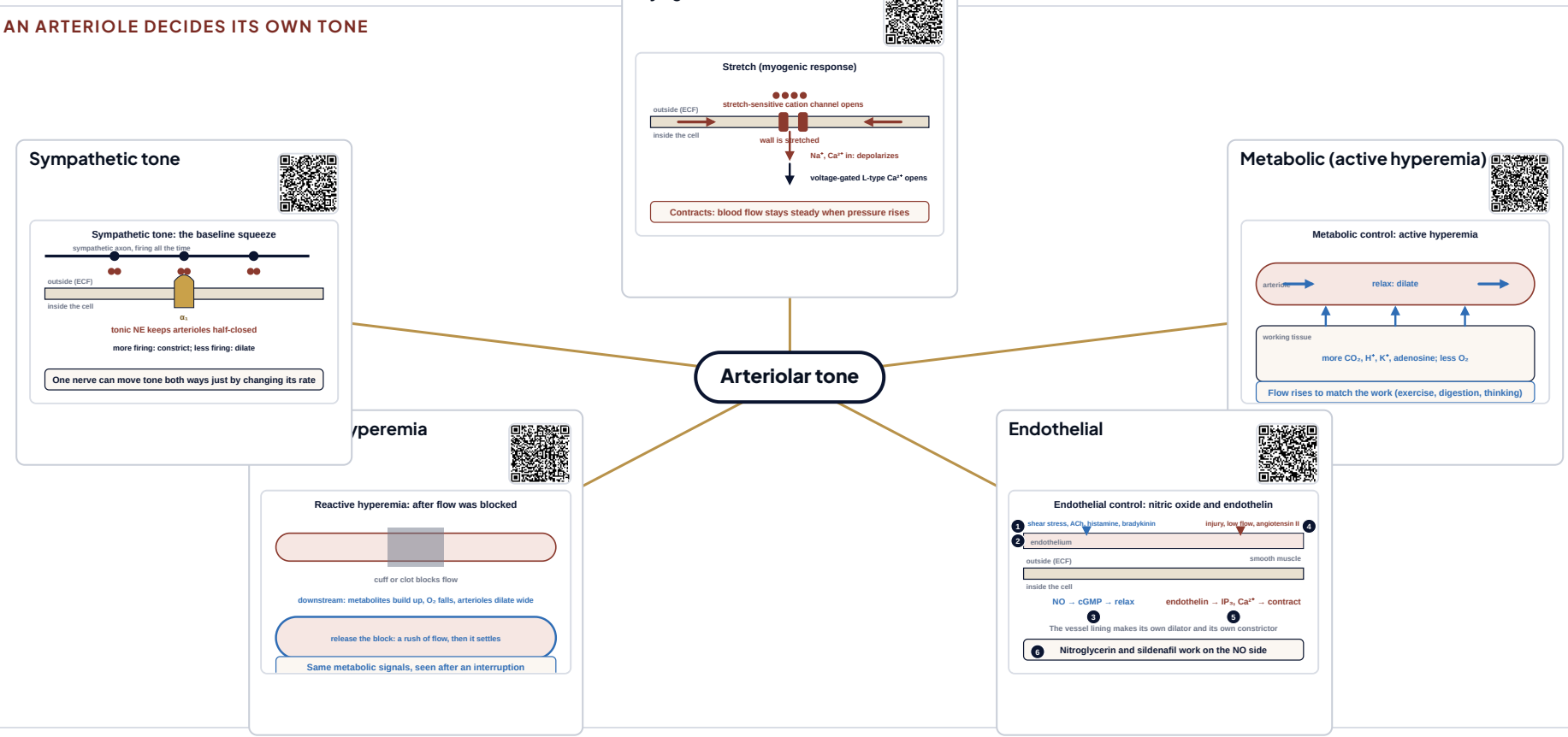
Name _____

C. How an arteriole decides its own tone

YOU SHOULD BE ABLE TO

Local control of blood flow. Explain myogenic, metabolic, and endothelial control of arte…se to increased tissue metabolism.

HOW AN ARTERIOLE DECIDES ITS OWN TONE



Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

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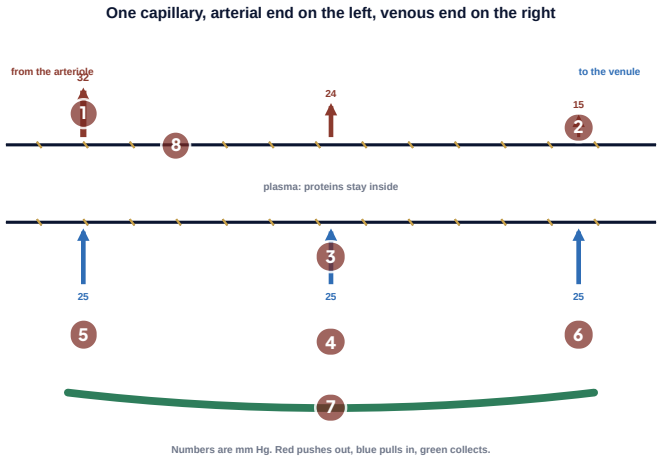
D. Capillary exchange, and edema

YOU SHOULD BE ABLE TO

Capillary exchange and Starling forces. Identify the four Starling forces at a capillary and determine the net direction of fluid movement at the arterial and venous ends.

Edema. Explain how increased capillary hydrostatic pressure, reduced plasma protein, increased permeability, or lymphatic blockage each produce edema and match a clinical scenario to its mechanism.

Lymphatic return. State the volume of fluid the lymphatic system returns each day and trace lymph from an interstitial space to the subclavian vein.



Starling forces along one capillary

1 _____ 5 _____
2 _____ 6 _____
3 _____ 7 _____
4 _____ 8 _____

WHAT YOU ARE LOOKING AT

One capillary opened out, blood flowing left to right. Red arrows: pressure pushing fluid out through the pores. Blue arrows: the pull of plasma proteins drawing fluid back in. Green: the lymphatic that collects what is left over.

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

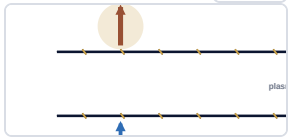
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Name _____

D. Capillary exchange, and edema (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

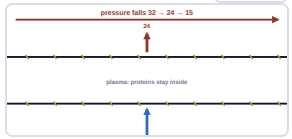
STEP 1



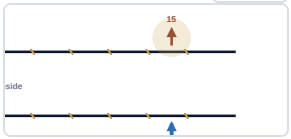
STEP 2



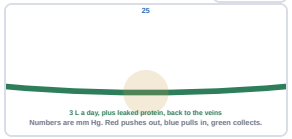
STEP 3



STEP 4



STEP 5



STEP 6



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

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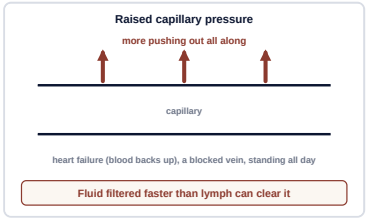
Name _____

D. Capillary exchange, and edema (continued)

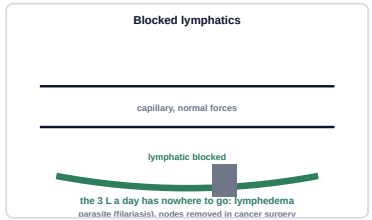
FOUR WAYS TO GET EDEMA

Edema

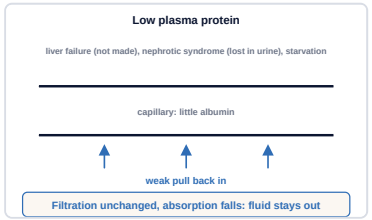
Raised capillary pressure



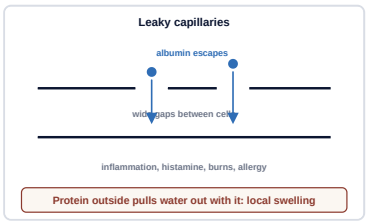
Blocked lymphatics



Low plasma protein



Leaky capillaries



LYMPH, BACK TO THE BLOOD: ABOUT 3 L A DAY

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

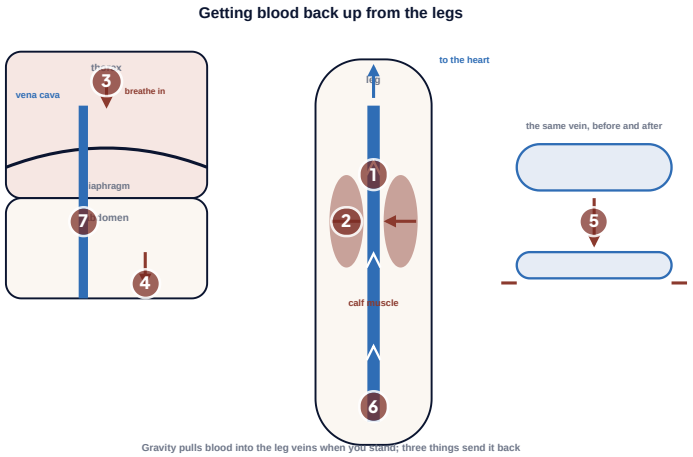
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Name _____

E. Venous return

YOU SHOULD BE ABLE TO

Venous return. Explain the skeletal muscle pump, the respiratory pump, and venoconstriction and predict the effect of prolonged standing on venous return and stroke volume.



Three pumps that bring blood back

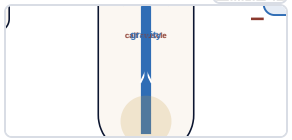
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WHAT YOU ARE LOOKING AT

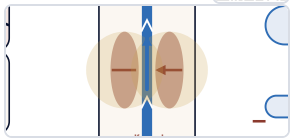
Left: the trunk with the diaphragm between thorax and abdomen and the vena cava running up through both. Middle: a leg vein with its valves and the calf muscle around it. Right: a vein before and after sympathetic constriction.

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

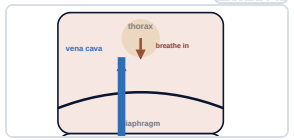
STEP 1



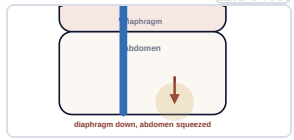
STEP 2



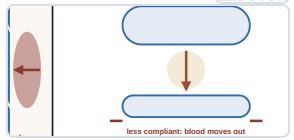
STEP 3



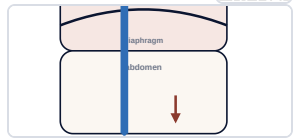
STEP 4



STEP 5



STEP 6



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

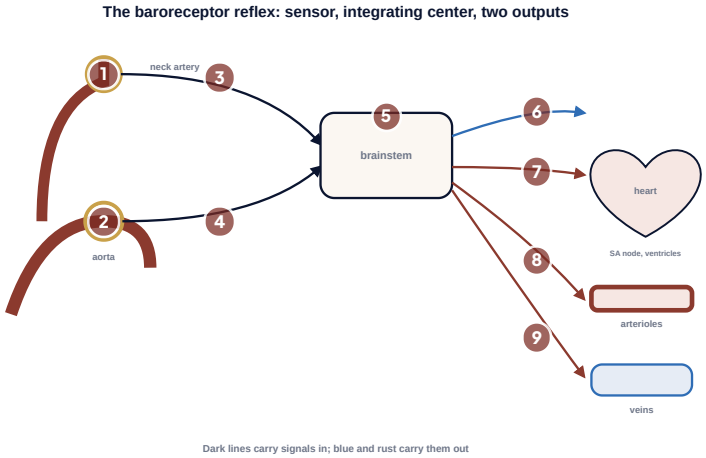
The vessels and the rules of flow, measuring a blood pressure, how arterioles and capillaries do their jobs, venous return, and the reflexes and hormones that hold pressure steady. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

F. Holding pressure steady: the reflex and the hormones

YOU SHOULD BE ABLE TO

Baroreceptor reflex. Trace the baroreceptor reflex from the carotid sinus and aortic arch through the medulla to the heart and vessels and state the response to a fall in blood pressure.
Hormonal control of blood pressure. Describe the roles of angiotensin II, aldosterone, antidiuretic hormone, and atrial natriuretic peptide in long term blood pressure control.
Adrenal cortex. Match the three cortical zones to aldosterone, cortisol, and the adrenal androgens and state the primary action of each.



The baroreceptor reflex

1 _____
2 _____
3 _____
4 _____
5 _____
6 _____
7 _____
8 _____
9 _____

WHAT YOU ARE LOOKING AT

Left: the two places pressure is sensed, the carotid sinus and the aortic arch. Middle: the medulla. Right: the three targets. Dark lines carry the signal in; blue is the parasympathetic output, rust the sympathetic output. Read it for a fall in pressure, such as standing up.

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

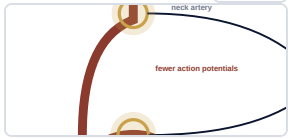
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Name _____

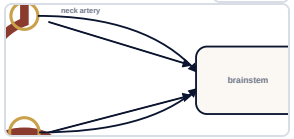
F. Holding pressure steady: the reflex and the hormones (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

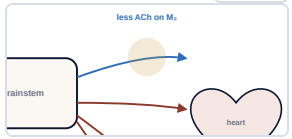
STEP 1



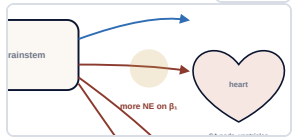
STEP 2



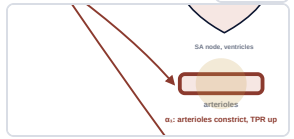
STEP 3



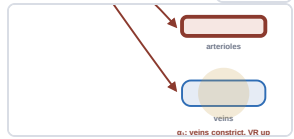
STEP 4



STEP 5

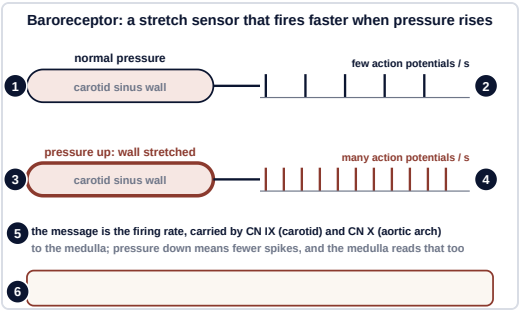


STEP 6



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

1 Carotid sinus baroreceptor: how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

Extra notes and drawings

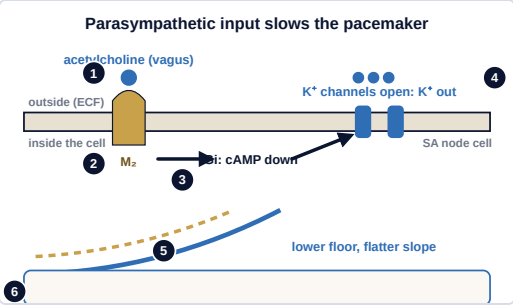
BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

The vessels and the rules of flow, measuring a blood pressure, how arterioles and capillaries do their jobs, venous return, and the reflexes and hormones that hold pressure steady. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

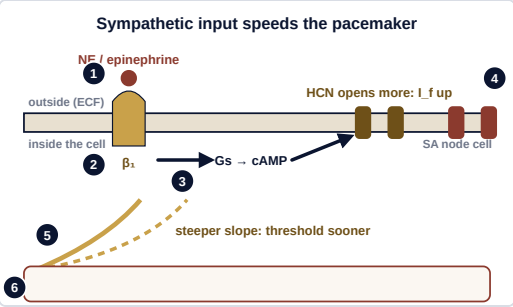
F. Holding pressure steady: the reflex and the hormones (continued)

6 Parasympathetic output to the SA node: how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

7 Sympathetic output to the heart: how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

Extra notes and drawings

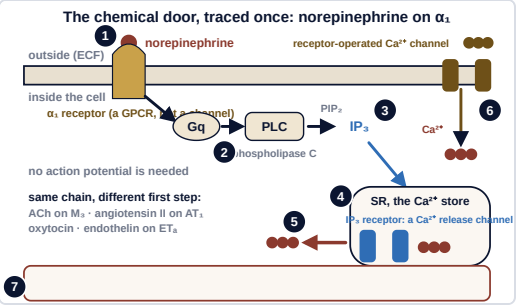
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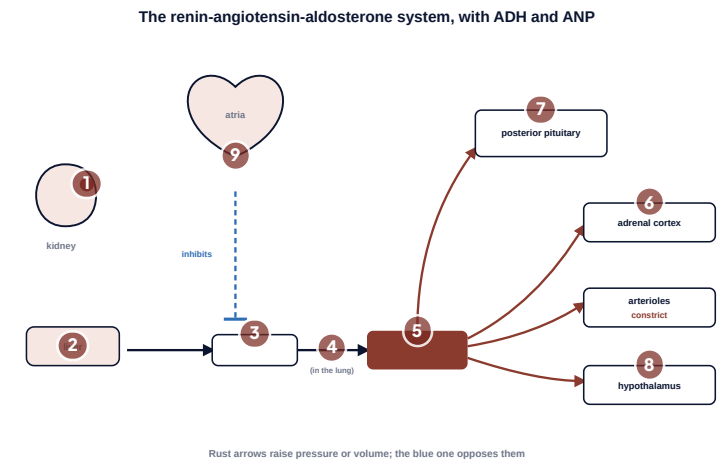
Name _____

F. Holding pressure steady: the reflex and the hormones (continued)

8 Sympathetic output to arterioles: how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.



The renin-angiotensin-aldosterone system, with ADH and ANP

1 _____
2 _____
3 _____
4 _____
5 _____
6 _____
7 _____
8 _____
9 _____

WHAT YOU ARE LOOKING AT

The long-term lever on blood pressure: a chain of hormones that ends in the kidney keeping salt and water. Follow the arrows from the kidney at the left, through the blood, to the targets on the right. ANP, in blue, pulls the other way.

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

The vessels and the rules of flow, measuring a blood pressure, how arterioles and capillaries do their jobs, venous return, and the reflexes and hormones that hold pressure steady. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

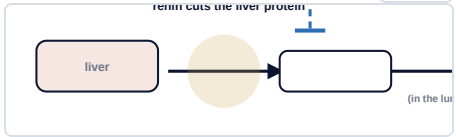
F. Holding pressure steady: the reflex and the hormones (continued)

HOW IT HAPPENS, IN ORDER (NUMBERS MATCH THE GOLD MARKERS ON THE SLIDE)

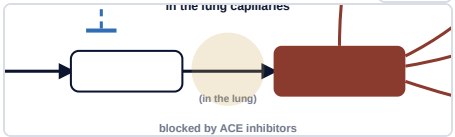
STEP 1



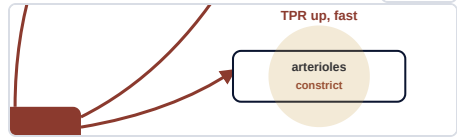
STEP 2



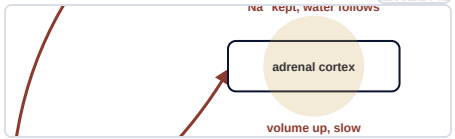
STEP 3



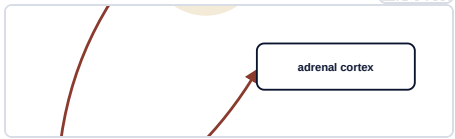
STEP 4



STEP 5



STEP 6



STEP 7



THE EFFECT: WHAT CHANGES BECAUSE OF THIS

Extra notes and drawings

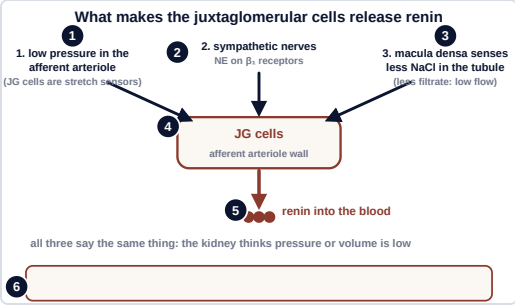
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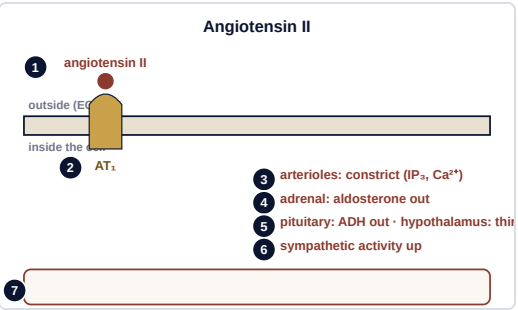
F. Holding pressure steady: the reflex and the hormones (continued)

1 Juxtaglomerular cells (renin): how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

5 Angiotensin II: how it works

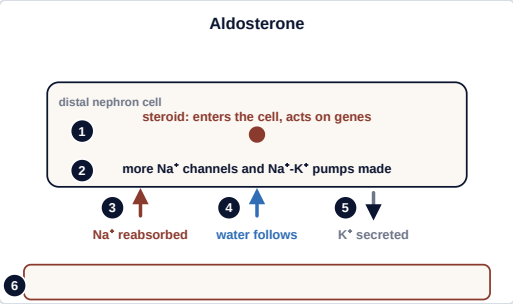


Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

Extra notes and drawings

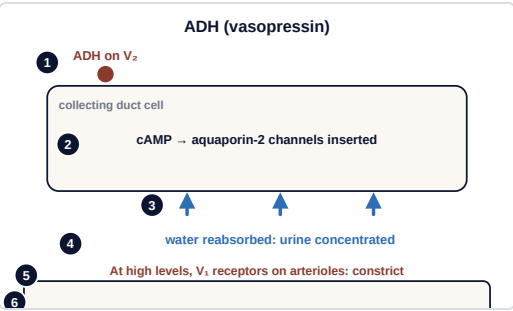
F. Holding pressure steady: the reflex and the hormones (continued)

6 Aldosterone (adrenal cortex): how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

7 Antidiuretic hormone (ADH, vasopressin): how it works

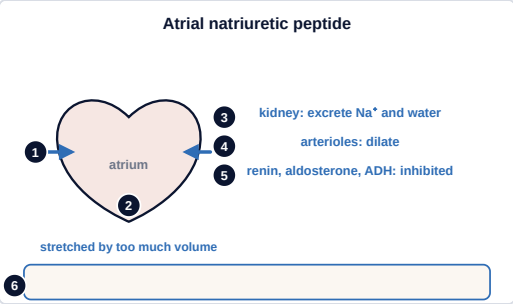


Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

Extra notes and drawings

F. Holding pressure steady: the reflex and the hormones (continued)

9 Atrial natriuretic peptide (ANP): how it works



Follow the numbers on the drawing and write what happens at each one, in your own words. Then fill the empty box at the bottom with the result.

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

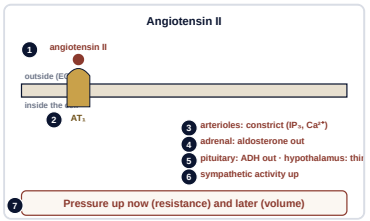
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Name _____

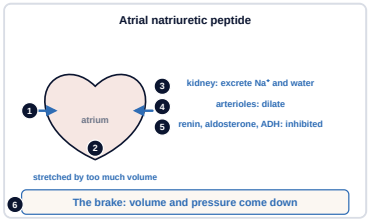
F. Holding pressure steady: the reflex and the hormones (continu

HORMONES THAT HOLD PRESSURE OVER DAYS

Angiotensin II

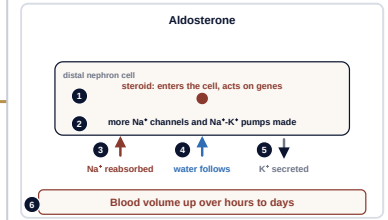


Atrial natriuretic peptide

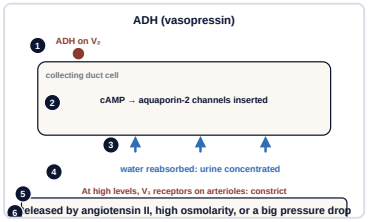


Blood volume

Aldosterone



ADH (vasopressin)



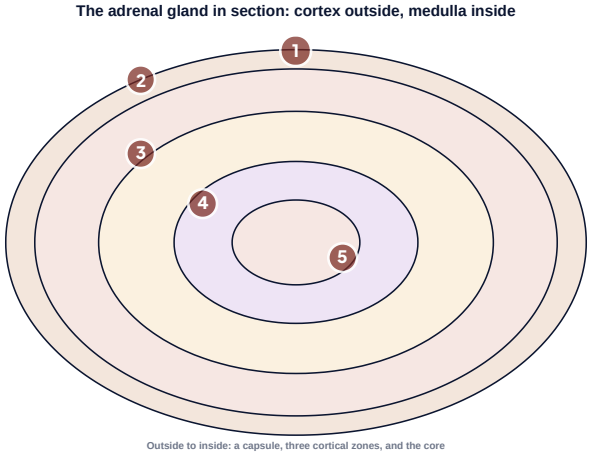
Extra notes and drawings

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Name _____

F. Holding pressure steady: the reflex and the hormones (continued)



The adrenal cortex in section

1. _____
2. _____
3. _____
4. _____
5. _____

THE ADRENAL CORTEX: ZONE, HORMONE, ACTION

	ZONA GLOMERULOSA	ZONA FASCICULATA	ZONA RETICULARIS
Hormone			
Controlled by			
Primary action			
Too much			

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

The vessels and the rules of flow, measuring a blood pressure, how arterioles and capillaries do their jobs, venous return, and the reflexes and hormones that hold pressure steady. Fill in as we go. Small square by a picture: scan it to see that picture full size on your phone.

Name _____

G. Two stress tests: exercise, and losing blood

YOU SHOULD BE ABLE TO

Cardiovascular response to exercise. Predict the change in heart rate, stroke volume, cardiac output, total peripheral resistance, and regional blood flow distribution during moderate exercise.

Compensation in hemorrhage and shock. Sequence the compensatory responses to blood loss and explain the point at which compensation fails.

MODERATE EXERCISE: WHAT CHANGES

	CHANGE	WHY
Heart rate		
Stroke volume		
Cardiac output		
Total peripheral resistance		
Mean arterial pressure		
Regional flow		

LOSING BLOOD: COMPENSATION, THEN FAILURE

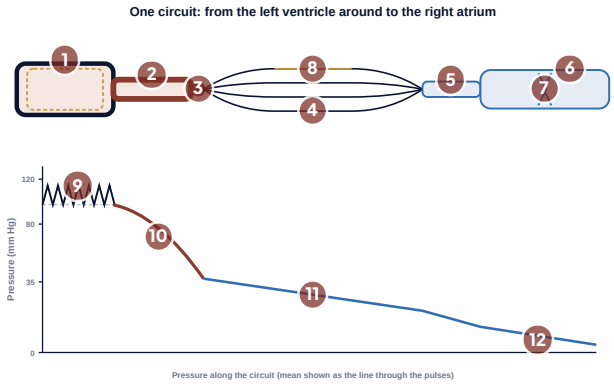
STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6

Extra notes and drawings

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure STRUCTURE NOTES

A. The vessels, and the rules of flow · Competency Vessel structure and function · Pressure flow and resistance · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____



One circuit, with the pressure in each vessel

1 _____

What it is

What it does

Why it matters

2 _____

What it is

What it does

Why it matters

3 _____

What it is

What it does

Why it matters

6 _____

What it is

What it does

Why it matters

7 _____

What it is

What it does

Why it matters

8 _____

What it is

What it does

Why it matters

9 _____

What it is

What it does

Why it matters

10 _____

What it is

What it does

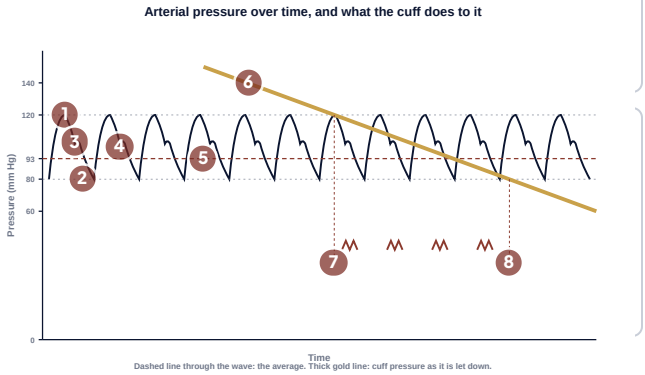
Why it matters

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

STRUCTURE NOTES

Arterial blood pressure, and how you measure it · Competency: Arterial blood pressure · Blood pressure measurement · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____



The arterial pressure wave and the cuff

1 _____

What it is

What it does

Why it matters

2 _____

What it is

What it does

Why it matters

3 _____

What it is

What it does

Why it matters

4 _____

What it is

What it does

Why it matters

5 _____

What it is

What it does

Why it matters

6 _____

What it is

What it does

Why it matters

7 _____

What it is

What it does

Why it matters

8 _____

What it is

What it does

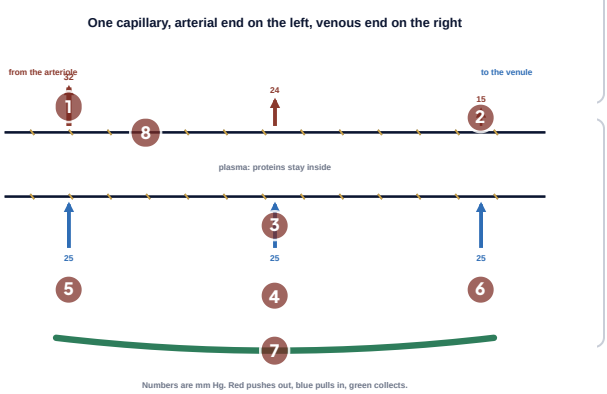
Why it matters

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure

STRUCTURE NOTES

Capillary exchange, and edema · Competency Capillary exchange and Starling forces · Edema · Lymphatic return · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____



Starling forces along one capillary

1 _____

What it is

What it does

Why it matters

2 _____

What it is

What it does

Why it matters

3 _____

What it is

What it does

Why it matters

6 _____

What it is

What it does

Why it matters

7 _____

What it is

What it does

Why it matters

8 _____

What it is

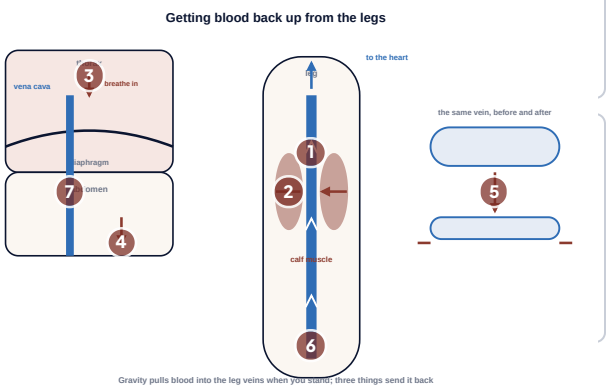
What it does

Why it matters

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure STRUCTURE NOTES

Venous return · Competency Venous return · one card per numbered part, the same three lines the slide reveals. Write the name, then what it is, what it does and why it matters, as we go.

Name _____



Three pumps that bring blood back

1 _____

What it is

What it does

Why it matters

2 _____

What it is

What it does

Why it matters

3 _____

What it is

What it does

Why it matters

6 _____

What it is

What it does

Why it matters

7 _____

What it is

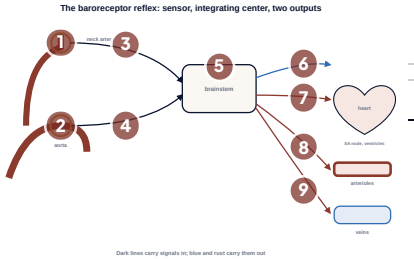
What it does

Why it matters

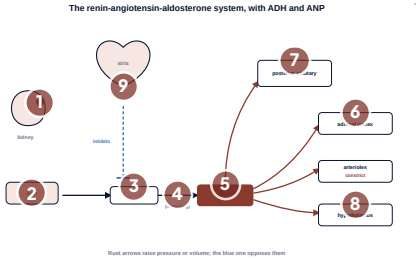
BIO 005 Week 10 · Pressure, Flow, and Blood Pressure STRUCTURE NOTES

Holding pressure steady: the reflex and the hormones · Competency Baroreceptor reflex · Hormonal control of blood pressure · Adrenal cortex · one card per numbered part, the same three lines the slide reveals · page 1 of 2. Write the name, then what it is, what it does and why it matters, as we go.

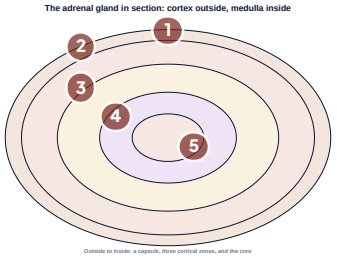
Name _____



The baroreceptor reflex



The renin-angiotensin-aldosterone system, with ADH and ANP



The adrenal cortex in section

1 _____
What it is

What it does

Why it matters

3 _____
What it is

What it does

Why it matters

1 _____
What it is

What it does

Why it matters

4 _____
What it is

What it does

Why it matters

2 _____
What it is

What it does

Why it matters

5 _____
What it is

What it does

Why it matters

3 _____
What it is

What it does

Why it matters

6 _____
What it is

What it does

Why it matters

4 _____
What it is

What it does

Why it matters

7 _____
What it is

What it does

Why it matters

5 _____
What it is

What it does

Why it matters

BIO 005 Week 10 · Pressure, Flow, and Blood Pressure STRUCTURE NOTES

Holding pressure steady: the reflex and the hormones · Competency Baroreceptor reflex · Hormonal control of blood pressure · Adrenal cortex · one card per numbered part, the same three lines the slide reveals · page 2 of 2. Write the name, then what it is, what it does and why it matters, as we go.

Name _____

8 _____

What it is

What it does

Why it matters

9 _____

What it is

What it does

Why it matters

1 _____

What it is

What it does

Why it matters

2 _____

What it is

What it does

Why it matters

3 _____

What it is

What it does

Why it matters

4 _____

What it is

What it does

Why it matters

9 _____

What it is

What it does

Why it matters

6 _____

What it is

What it does

Why it matters

7 _____

What it is

What it does

Why it matters

4 _____

What it is

What it does

Why it matters

5 _____

What it is

What it does

Why it matters